

The economics of risk, uncertainty and learning in the adoption of new agricultural technologies: where are we on the learning curve?

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Abstract

The roles of risk, uncertainty and learning in the adoption of new technologies are reviewed. Although they have been emphasized in many commentaries about the adoption process, they have been directly addressed in only a minority of the large literature relating to the adoption of innovations. Risk, uncertainty and learning play a number of distinct roles in the process of adopting new technologies. These distinct roles have often been blurred or treated incompletely in past research. Theoretical and empirical literature, which explores and evaluates these various roles is reviewed, with a focus on agricultural technologies. A conceptual framework that captures the main impacts and roles of risk is outlined. A range of research needs and emerging issues for risk and technology adoption in agriculture are discussed. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Risk has often been considered to be a major factor reducing the rate of adoption of a new technology (e.g. Lindner et al., 1982; Lindner, 1987; Tsur et al., 1990; Leathers and Smale, 1992; Feder and Umali, 1993). However the issue of risk in adoption has rarely been addressed adequately, and strong empirical evidence to test

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the common view about its importance and impact has been rare and scattered. In this paper we review the available theoretical and empirical literature for agriculture to attempt to present the state of current knowledge in this important area.

We identify a number of aspects of risk, uncertainty and learning which should be considered as distinct elements of the problem, but which are often either blurred together or treated incompletely in the literature. The core elements are:

- learning which improves the farmer's ability to implement the new technology,
- learning which allows the farmer to make better decisions about the new technology,
- perceptions of the farmer about the present and future probability distribution of economic returns from the new technology,
- perceptions of the farmer about the covariance of economic returns between new and old technologies,
- the strength and direction of risk attitudes of the farmer (i.e. risk averse, risk neutral, risk preferring),
- and the option value from delaying where there are fixed costs of adoption.

The paper proceeds as follows. In the next section, we outline findings from a number of past literature reviews on adoption and diffusion of new technologies. The subsequent section focuses more narrowly on the minority of that literature which has explicitly addressed issues of risk, uncertainty and learning. The fourth section contains a conceptual framework for considering these issues, drawing on key insights from the literature. The final section examines a number of research needs and emerging issues in risk and technology adoption.

2. Reviews of technology adoption literature

2.1. Contribution of different disciplines

Reviews of adoption and diffusion studies with an historical perspective can be found in Rosenberg (1976), Lindner and Pardey (1979), Lindner (1987), Grubler (1992), Ruttan (1996) and Sarkar (1998). Sociologists and economists have been the main contributors to the conceptualization of the adoption decision process at the individual firm level. Sociologists have traditionally concentrated on the distinguishing characteristics of the potential adopters and opinion leaders, prospective adopter's perception of the attributes of innovations, rates of adoption of new technologies and ideas, and finally the communication channels involved in the different stages of the innovation-decision process. Ruttan (1996) commented that a positive aspect of the limited, but valuable, convergence of the application of methodologies from economics and sociology has been the identification of the learning behavior of individuals as an important factor in the adoption process.

Griliches' research on hybrid corn in the USA was one of the first economic studies of adoption of rural innovations by economists (Griliches, 1957, 1960).

Although not surprising now, his conclusion that economic variables were the major determinants of technological change and adoption of innovations was pathbreaking at the time. Mansfield (1961), who conducted contemporaneous empirical studies of adoption rates of a number of industrial innovations, also concluded that the adoption of innovations is determined by their economic attributes.

Rosenberg (1976), in a review of research on adoption of innovations, suggested that poor explanatory power of models put forward by sociologists are a consequence of inadequate attention paid to the role of economic variables. He argued that “practical businessmen tend to remember what social scientists often forget: that the very rapidity of the overall pace of technological improvement may make a postponed adoption decision privately (and perhaps even socially) optimal” (Rosenberg, 1976, p. 535). He added,

decisions to postpone the adoption of an innovation are often based upon well-founded and insufficiently-appreciated expectations concerning the future time-flow of further improvements. Even the most widely accepted justification for postponement, the elimination of conspicuous but not overwhelmingly serious technical difficulties, or ‘bugs’, can reasonably be interpreted as merely a special case of expectations of future technological improvement (p. 534).

He suggested that entrepreneurs may be making appraisals of the future returns to innovations of greater objective validity than are made by social scientists who invoke all sorts of extra-rational factors to account for the delay or “lag” in the adoption and diffusion of innovation.

Ruttan (1996) observed that sociology, and rural sociology in particular, had lost its dominant role in research on the adoption and diffusion of innovations by the mid 1980s. He suggested that this was due to the over reliance by sociologists on survey research methodology as their main analytical tool. He argued that they failed to embrace the more formal analytical methods, such as behavioral modeling. Ruttan pointed out some studies by geographers, economists and technologists that used simulation models to understand the processes of technological innovation, substitution, replacement and spatial diffusion.

Ruttan reviewed a suite of studies that used what he referred to as equilibrium models of diffusion. He noted that in these models, diffusion is conceptualized as a transition between equilibrium levels, influenced by changing economic attributes (e.g. prices) and a changing environment (e.g. crop yields). In these models, diffusion does not involve learning. Rather, diffusion occurs as a result of the interaction of changes in the innovation and the adoption environment. Therefore the process of technological change is one of substitution rather than a diffusion phenomenon. Ruttan saw this type of conceptualization of the diffusion process as a further adaptation of sociological diffusion models which some marketing economists have used to forecast consumer demand. He commented that the more radical applications of the equilibrium approach completely abandon the communication model severing any intellectual links with the sociological origins of diffusion research.

Lindner and Pardey (1979) noted that the study of spatial diffusion of innovations by social geographers has made a significant contribution to the understanding of

the diffusion process in an area ignored by economists and sociologists: the role which infra-structure and/or supply aspects play. Although spatial diffusion models have attempted to describe the aggregate spread of an innovation, they have added little to our understanding of the innovation-decision process. They fail to account for the influence of other adopters, the influence of agents supplying the innovation and the influence of personal experience and experimenting with the innovation. It appears that the conclusions and criticisms of Lindner and Pardey's (1979) review remained unheeded in most of subsequent studies of the 1980s and the 1990s (e.g. Saha et al., 1994).

Lindner and Pardey noted that many of the past temporal and spatial diffusion models put forward by economists were based on a general assumption that the potential adopter is a passive participant in the adoption process. Spatial diffusion studies have shown two strands of thought. One strand is concerned with the neighborhood effect and resistance, while the other strand focuses on the role of technology suppliers and innovators. In general, spatial and temporal models have been concerned with the spread of knowledge about the innovation, not the rate of adoption of the innovation itself. This means they have ignored the central determinant of the rate and pattern of adoption which is the decision process involved in moving from a state of awareness to actual adoption.

2.2. Different types of empirical adoption studies

Lindner's (1987) survey of research on adoption at the firm level found that most studies were concerned with identifying those characteristics of the adopters or the innovations that influenced adoption decisions. He classified these empirical studies into two categories. The first were the cross-sectional studies in which the main question was why do some producers adopt an innovation while others reject it. The majority of the empirical studies of adoption are concentrated in this category. Recent examples include Shapiro et al. (1992), Smale et al. (1994), Bosch et al. (1995), Nkonya et al. (1997) and Marra et al. (in press). The key features of these cross-sectional studies that are relevant to this study will be discussed in more detail in the section dealing with risk and learning. The second type of adoption studies is the temporal studies that are concerned with the determinants of the timing of adoption. These studies typically try to identify why some producers are early adopters while others are laggards. There are fewer of these studies mainly due to the difficulties in the collection of data that can be used for this type of analysis. Examples include O'Mara (1971), Lindner et al. (1982), Feder and Slade (1984), Lindner and Gibbs (1990), Foster and Rosenzweig (1995), Carletto et al. (1996), de la Briere (1996), Burton et al. (1998) and McWilliams et al. (1998).

3. The roles of risk, uncertainty and learning in the adoption process

Research on the economics of technology adoption under uncertainty also has taken two paths. One path considered technology adoption from the standpoint of

investment in a durable asset with an uncertain future value. The other explored the relationship between the riskiness of the technology and the utility of a risk-averse decision maker. Both are reviewed below.

3.1. The theory of technology adoption under risk or uncertainty

Around the same time as the Rosenberg review, environmental and resource economists began to borrow several pertinent notions from the field of finance, namely, that uncertainty about the future value of an investment and its sunk costs provide an alternative explanation for investment lags (Arrow and Fisher, 1974). The thrust of this research was to show that there is an *option value* of waiting to invest when there is uncertainty about the future payoffs and sunk costs in the sense that one cannot recoup all investment costs when disinvesting (also called the degree of irreversibility of the investment). A series of papers in the economics literature developed these notions as they apply to the investment decisions of extractive industries (see Dixit and Pindyck, 1994, for a comprehensive review and synthesis of this literature). Only recently has this line of inquiry been applied to agricultural investment problems (Chavas, 1994; Purvis et al., 1995; Zhao, 2000). Chavas argues that, because of sunk costs, it may be socially optimal for government-provided price floors to reduce the uncertainty of the investment. Purvis et al. applied this idea to dairy farmer investment in new waste management technology and found that, compared to the net present value approach to the investment decision, the option value approach implied a significantly higher income stream was required before investment would take place. Zhao, using a game-theoretic approach, considered the case where the option value of waiting to adopt is related to the opportunity to observe earlier adopters' experience with the technology.

Just and Zilberman (1983) were among the first to propose a theory of technology adoption under uncertainty using the expected utility framework. The Just and Zilberman model is an extension of the original Baron (1970) and Sandmo (1971) expected utility approach to producer behavior under uncertainty. Their approach provides a theoretical basis for study of the role played by firm size, risk attitudes and the joint distribution of returns, credit constraints, and fixed costs of adoption in the choice between two risky technologies. One theoretical result of their model is that when the covariance of returns between the old and new technologies is high and the decrease in absolute risk aversion as wealth increases is sufficiently large, there may be a limit to the proportion of a farmer's cropland devoted to a new, profitable technology. They also consider the effect on adoption of the fixed costs of the new technology.

3.2. Empirical studies accounting for risk and uncertainty

There is a general scarcity of empirical studies that have adequately addressed the role of risk and uncertainty in adoption. Feder et al. (1985) in their review of the adoption literature attributed this scarcity to difficulties in observing and measuring risk and uncertainty, as noted by Lindner et al. (1982) and Akinola (1986).

Smith and Mandac (1995) estimated farmers' subjective probabilities of the distribution of yields through a direct elicitation technique. They then compared these estimated probability distributions to relatively objective estimates. Their results showed that while the means of subjective and objective distributions were similar, farmers seriously underestimated variances when compared to more accurate estimates of objective variances.

The rare attempts to seriously investigate empirically the roles of risk and uncertainty in adoption include studies by O'Mara (1971, 1980), Binswanger et al. (1980), Herath et al. (1982), Byerlee and Hesse de Polanco (1986), Marra and Carlson (1987), Kebede (1992), Shapiro et al. (1992), Smale et al. (1994), Goodwin and Kastens (1996) and Abadi Ghadim (2000). Even fewer studies of risk and adoption have used direct interview techniques to investigate the effect of farmers' risk attitudes and perception of riskiness of enterprises on their allocative decisions (Binswanger, 1980; O'Mara, 1983; Lindner and Gibbs, 1990; Smale et al., 1994; Huirne et al., 1997; Abadi Ghadim, 2000). However, with the exception of Abadi Ghadim (2000), these studies have generally had low explanatory power.

O'Mara (1980) studied the effect of perceived riskiness of new varieties of grains on the adoption decisions of Mexican farmers. To do this he obtained farmers' subjective yield distributions for the new crops. He found that farmers were modifying these subjective distributions over time as they acquired more information. These subjective estimates of the riskiness of crops were found to influence the actual adoption decisions of farmers. Smale et al. (1994) included the testing of perceived risk and subsistence ratio as one of several explanations for why farmers grow more than one variety of the same crop.

Binswanger et al. (1980) elicited the risk preferences of a sample of Indian farmers. They used several elicitation techniques, one of which included gambling questions with real monetary pay-offs. These methods measured farmers' levels of risk aversion, which were then used in regression analyses of farmers' adoption decisions. Statistical significance tests showed mixed results, and their findings were inconclusive with regards to risk aversion.

Byerlee and Hesse de Polanco (1986) analyzed farm survey data from Mexico to investigate the reasons for stepwise adoption of components of packages of practices. They hypothesized that the time of initiation of adoption and the rate of adoption would be functions of (a) profitability, (b) riskiness, (c) divisibility, (d) complexity, (e) availability and (f) interactions between components of the innovation. Their regression model showed that adoption of each innovation was explained primarily by its profitability and riskiness.

Marra and Carlson (1987) provided an empirical test of some of Just and Zilberman's theoretical results using farm-level data on adoption of double-cropped wheat/soybeans. They found evidence to support the proposition that the combined effects of decreasing absolute risk aversion and covariance of returns are likely to be limiting factors in the farm size–adoption relationship.

Smale and Heisey (1993) investigated the reason why farmers experiment with recommendations about an innovation and often adopt parts of the technological package associated with the innovation in a step-wise manner rather than adopting

the whole package at once. They conceptualized the adoption of a new agronomic package, consisting of a new crop that also required adoption of an alternative fertilizer, as a system of simultaneous input-choice equations. They found that simultaneous equation estimation produced different levels of predicted adoption probabilities than did single equation estimation. They concluded that it is important to consider and account for the relationships between the components of the innovation package.

Kebede (1992) found that the adoption of new technologies by Ethiopian farmers was significantly related to their degree of risk aversion. They reported that income, farm size, education, family labor and experience were negatively related to risk-taking behavior in farming areas where off-farm income diversification opportunities existed. These factors were found to have the opposite effect in areas where farming was the only source of income.

Marra and Carlson (1990) studied the effect of relative risk and risk aversion on the allocation of acreage between full-season soybeans and double-cropped wheat/soybeans in the southeastern USA. The expected utility framework used in conjunction with a covariance decomposition called Stein's theorem allowed direct estimation of the Arrow–Pratt risk aversion coefficients without first assuming a functional form for the utility function. They found that, with state-level data, the relative riskiness of the two production techniques and risk aversion mattered somewhat in most cases in the aggregate decision to double crop.

Shapiro et al. (1992) used a Tobit model to explain the effect of several variables including risk aversion also on adoption of double cropping in the USA. This was one of the rare studies where the variables for risk preferences and riskiness were quantitative estimates (rather than relatively weak proxy variables) based on a rigorous conceptual framework of risk. The producer's risk preference was measured by a Pratt–Arrow measure of risk attitude elicited using the method reported in King and Robison (1981). They found a wide range of risk preferences in the sample population of farmers; 45% were risk averse, 16 percent were risk neutral and 39% risk preferring. They elicited farmers' perception of riskiness of enterprises by asking them to construct frequency distributions using the fixed interval approach (Raiffa, 1968). This required the farmer to consider the yields and prices they would expect in the subsequent 10 years. They found that adopters were, on average, more risk averse than non-adopters. However, other factors, including differences in their risk perceptions, were more important than their risk preferences in explaining adoption. Their results showed that adopters perceive double cropping to lower risk and increase total crop revenues.

The findings of the Shapiro study, showing higher levels of adoption of double cropping with higher risk aversion, contrasts with the findings of Brink and McCarl (1978), who found the opposite relationship for a similar sample of farmers, and with the Marra and Carlson (1990) study. Shapiro et al. (1992) argued that these contradictory results could be explained by noting that this sample of farmers displayed a wide range of risk preferences. They suggested that this may be “related to Young's contention that risk preferences differ by situation and level of risk” (Young, 1979, p. 39)

Smale et al. (1994) found that Malawian maize growers' perceptions of the relative riskiness of new seed varieties influenced the probability of their adoption and intensity of cultivation. These researchers elicited subjective yield distributions from individual farmers for the different varieties of maize to calculate relative riskiness estimates and also included in their analysis variables other than risk aversion that could explain the adoption patterns.

Finally, Abadi Ghadim (2000) conducted a study that comes close to implementing and estimating a complete set of risk impacts related to adoption. The study involves a 3-year series of personal interviews with 130 crop producers in Western Australia, collecting information on their actual and planned adoption behavior for a new crop type, chickpeas. Subjective probability distributions for yields and prices were collected for the new crop and traditional land uses. In addition, subjective beliefs about the covariance between the yields of chickpea and traditional crops were elicited. Arrow–Pratt coefficients of risk aversion were elicited based on a set of questions related to hedging. Changes in subjective perceptions from year to year were observed and related to the farmers' experiences in trials of the new crop. Three limited dependent variable models, Tobit, Probit and Heckman, were used to estimate the empirical model. Each gave a high degree of goodness-of-fit. Results showed that some determinants of the decision to adopt the innovation are different to those that determine the decision regarding the intensity of adoption. The study provided strong empirical evidence to support the primarily economic character of the adoption decision and also highlighted the importance of economic risk in the adoption process. The two risk-related factors with greatest impact on the adoption decision were risk aversion and relative riskiness. Risk aversion tended to reduce adoption, and to do so to a greater extent as relative riskiness and scale increased.

3.3. The role of information and learning

To model the micro-level adoption process from a dynamic (temporal) perspective it is necessary to treat adoption as a process involving acquisition of information and learning. A survey of recent literature on the dynamics of the adoption innovation shows two distinct approaches to modeling this process. The first is to model the adoption decision of individuals over time by the inter-temporal changes in certain explanatory variables. For innovations that are “divisible” (e.g. a new crop) and can be adopted in a stepwise manner, the adoption decision also involves a decision regarding the intensity of adoption at any given time period along the adoption time path. This “snapshot” of the adoption process is the subject of the second approach.

Warner (1974) concluded that learning and imitation are central to the adoption story. He interpreted the specification of the logistic function for the diffusion of an innovation over time as implying that potential adopters initially have a cautious approach toward adopting the innovation. Where possible, they initially experiment with the innovation on a trial basis. Before deciding whether to adopt or not, they seek information on the cost and value of the innovation from their own and other users' experiments. As more information is gathered, the decision-makers are able to

increase their knowledge about the overall attractiveness of the innovation through improvement in their knowledge of the best use of the innovation and reduction in their uncertainty about its potential benefits. This in turn increases the pace of adoption. He asserted that the nature of learning and imitation has been embedded in the mathematical diffusion model both by assumption and by derivation. He argued that there was strong empirical evidence for the hypothesis that efficiency in the use of a novel technology increases with experience (learning-by-doing). Lindner and Pardey's (1979) conclusions supported this proposition.

Lindner (1987) argued that a considerable amount of firm-level adoption behavior could be explained through *correct* specification of “informational” variables. In most of the adoption studies, proxies are used to represent informational variables. These typically include proxies for the cost of acquiring information and proxies for the incentive to acquire information. Examples of proxies for the cost of acquiring information includes education level, distance to nearest current adopter, and availability of extension services. Farm size is an example of a proxy that has been used for the incentive to acquire. Recent studies involving such proxies include Nowak (1987), Harper et al. (1990), Sinden and King (1990), McNamara et al. (1991), Goodwin and Schroeder (1994), Saha et al. (1994), Ahouissoussi (1995), Eckman et al. (1996), Goodwin and Kastens (1996) and Bhattacharyya et al. (1997).

Lindner and Gibbs (1990) is one of the few empirical attempts at studying how farmers collect innovation-specific information from on and off-farm sources and revise their beliefs about it. Due to their small sample (20) and problems with their elicitation techniques, they could not confirm nor reject whether farmers revise their subjective beliefs about new varieties in a Bayesian manner.

Fischer et al. (1996) is another good example where the role of quality of information on the adoption of wheat varieties was investigated. They used empirical data collected from a sample population of 121 farmers over 3 consecutive years. This data set contained elicited subjective yield distributions of six wheat varieties from a subset of 50 farmers. From these distributions they computed the farmers' subjective beliefs about the mean yield and the variance of the yield of each wheat variety for each year. For comparison purposes they also computed similar moments of the distribution of the yields of the same wheat varieties from historical data of three experimental trial sites. Fischer et al. used these data in their Bayesian “random-effects” model. In their model not all pieces of information added equally to the potential adopter's knowledge about the innovation. Their model indicated that “effective information” is generated much more slowly than in previous models (Feder and O'Mara, 1982). It also suggested that the amount of effective information is well approximated by the number of years since the trialing of a particular wheat variety began. They concluded that simply the amount of information available about an innovation might not be a good indication of effective information.

The results of Abadi Ghadim (2000) also highlighted the key role played by trialing to provide information that reduces uncertainty and promotes skill development. He was able to obtain significant results for separate variables related to the two aspects of learning from trials. The covariance of new and old crop performance was highly significant, and was interpreted as reflecting the value of information to

reduce uncertainty. This is because a high covariance allows farmers to learn more about the yield distribution of the new crop by relating its trial yields to yields of other familiar crops. A subjective rating by farmers of the value of information from trialing chickpeas versus trialing a new variety of their major crop, wheat, was also highly significant, probably for the same reason. The level of past farmer experience in growing the new crop was positively related to continued adoption, probably reflecting in part the importance of skill development. Finally there was a significant effect on adoption of a variable representing the perceived likely difference between crop yields before and after skill development following experience in producing the new crop.

Marra et al. (in press) provide a new empirical test of the hypotheses put forth in Fischer et al. as to what sort of information is more important (effective) in the individual adoption decision. They consider different sources of direct profitability estimates, ranging from own-farm trials to state level averages, as well as indirect information such as how “popular” the new technology is within a relevant area. They compare the Fischer et al. proposition of effective information to a competing hypothesis put forth by Ellison and Fudenberg (1993). This model allows for popularity weighting of information by assuming potential adopters take account of their neighbors’ decisions in making their own. Marra et al. tested the hypotheses using as their empirical example the adoption of transgenic cotton in 1996 and 1997 in the southeastern USA. Assuming a random utility model and using farm-level survey data, their results show strong support for the Fischer et al. hypothesis of “effective information” and weaker support for the “popularity” hypothesis.

4. A conceptual framework

Abadi Ghadim and Pannell (1999) presented a framework that conceptualizes adoption as a multi-stage decision process involving information acquisition and learning-by-doing by growers who vary in their risk preferences and their perceptions of an innovation’s riskiness. As their framework captures the key issues highlighted in the foregoing literature review, we present it here in outline form as a summary of the current state of thinking about risk, uncertainty, learning and adoption of innovations.

The framework is designed around adoption of a divisible technology, with a new crop type (chickpeas) used as the example. The adoption decision is about whether the new crop should replace the traditional crop and, if so, to what extent.

4.1. Learning leads to skill improvement

As noted earlier, Lindner and Pardey (1979) emphasized the importance of personal experience and experimentation in the adoption process, and Abadi Ghadim (2000) highlighted skill improvement as one important consequence of experience. In essence, skill improvement need not be considered in a stochastic framework. Abadi Ghadim and Pannell (1999) derive the following equation to address the

question of whether income from trialing the new crop in year one plus the value of improving the farmer's skill in growing it in future years outweighs the loss of income from foregoing the alternative (traditional) crop.

$$\Pi^* - \Pi^0 = (G_{c1}^* - \bar{G}_A) \cdot A_{c1}^* + I_S \quad (1)$$

where Π^* is the net present value of income if the new crop is trialed at the optimal level in year 1, and Π^0 is the net present value of income if the new crop is not trialed in year 1. Meanings of other symbols used here and in subsequent equations are given in Table 1. I_S represents the difference between NPV of profits for years subsequent to year one. I_S is a monetary value that arises from the improvement in the farmer's skills at growing the crop due to experience and information learned in a trial in year one. It is a value of information which differs from that usually discussed in the decision theory literature (e.g. Anderson et al., 1977). The value arises from changing the technical parameters of the production function, rather than from better decision making. It encompasses any adjustment in area of the new crop and the alternative crop in future years as a result of the farmer's higher skill level after the first year. Abadi Ghadim and Pannell (1999) showed that I_S could be decomposed into two elements as follows:

$$I_S = \text{NPV}_{t=2}^n [(\bar{G}_{ct}^* - \bar{G}_{ct}^{**}) \cdot A_{ct}^{**} + (\bar{G}_{ct}^* - \bar{G}_A) \cdot \Delta_{Ac}] \quad (2)$$

Table 1
Glossary of the variable names

Variable	Description of the variable name
G_c	Gross margin of the innovative enterprise
G_A	Gross margin of the alternative (traditional) enterprise
A_T	Total resources available (total area of the farm)
A_c	Resources allocated to the innovation (area of chick peas)
$\bar{G}_c$	Mean gross margin of chick peas over the area planted
$\bar{G}_A$	Mean gross margin of the alternative enterprise over the area planted
A_{ct}^*	Optimal allocation of resources to the innovation in season t if the farmer trials the innovation in the first year
A_{ct}^{**}	Optimal allocation of resources to the innovation in season t if the farmer does not trial the innovation in the first year
$\bar{G}_{ct}^*$	Gross margin of the innovation if the farmer uses A_{ct}^* as the planting rule.
$\bar{G}_{ct}^{**}$	Gross margin of the innovation if the farmer uses A_{ct}^{**} as the planting rule.
t	Time in yearly increments
n	Number of years in the farmer's planning horizon
I_S	Value of information from trialing for skill development
I_D	Value of information from trialing for decision making
Δ_{At}	Change in the allocation of resources to the innovation in year t as result of a trial in year 1
$\text{NPV}_{t=2}^n$	Net present value of the profits from year 2 to year n
E	Expected value
U	Utility

The two elements of I_S are: the gain in profitability on the area which would have been sown to the new crop in future years even without it being grown in year one, $(\bar{G}_{ct}^* - \bar{G}_{ct}^{**}) \cdot A_{ct}^{**}$, plus the gain in profit on the area converted from the traditional crop to the new crop in future years as a result of trialing the new crop in year one, $(\bar{G}_{ct}^* - \bar{G}_A) \cdot \Delta_{Ac}$. One might hypothesize that I_S would increase at a decreasing rate with an increasing scale of the trial. Larger trials would provide more opportunities for skill development, but as scale increased the marginal benefits in terms of skill development would probably diminish.

4.2. Learning reduces uncertainty and improves decision making

The second outcome of the trialing and experimentation process emphasized by Lindner and Pardey (1979) and Lindner (1987) is improved information about the performance of the innovation. This, then, introduces uncertainty to the model, with trials providing an opportunity to reduce uncertainty. Assume initially that the farmer's objective is to maximise the expected net present value of profits. Before conducting the trial, the farmer has a subjective perception of the possible values of G_c , the gross margin of the new crop on each unit of his or her land. For a given area of the new crop, the gross margin varies from acre to acre and the mean over the area is denoted by $\bar{G}_c$. $\bar{G}_c$ varies according to the area of the new crop grown. Now, the farmer is uncertain about the value of $\bar{G}_c$, but is able to subjectively state a probability distribution for it. Given the farmer's objective to maximise expected NPV, $E(\bar{G}_c)$ would be the value used in a standard decision theory model to represent the pay-off from the new crop in each of the years. The E operator in $E(\bar{G}_c)$ signifies the expectation over uncertain states of nature, while $\bar{G}_c$ signifies the mean across all the area devoted to the new crop.

From the information generated in the trial, the farmer revises his or her subjective beliefs about the profitability of the new crop. Based on this revised (hopefully more accurate) perception, the farmer decides whether or not to continue growing the new crop and, if so, what area of the farm to devote to it.

A trial in year t provides information that allows improved estimates of $\bar{G}_{ct}$ for subsequent years. This in turn allows improved selection of A_{ct} for subsequent years. The gain in expected profit $E(\bar{G}_{ct})$ as a result of the changes in A_{ct} constitutes the value of improved decision making from the trial, I_D .

It should be clear that I_D is different from I_S but the two interact because both are related to changes in the area of the innovation in later seasons. In the case of I_S , improvements in G_c encourage increases in A_c , while for I_D , better knowledge of the crop's performance may either increase or decrease the area selected to be grown.

Mathematically, including I_D in the decision of whether to trial chick peas in the coming year (i.e. whether $A_{c1}^* > 0$) gives

$$\Pi^* - \Pi^0 = \bar{G}_{c1} \cdot A_{c1}^* - \bar{G}_A \cdot A_{c1}^* + I_S + I_D \quad (3)$$

I_D can be reduced to:

$$I_D = \text{NPV}_{t=2}^n [(\bar{G}_{ct}^* - \bar{G}_A) \cdot \Delta_{Ac}] \quad (4)$$

Thus, the value of information from trialing in the model is the gain in profit on the area converted from the traditional crop to the new crop in later years as a result of the trial.

Like I_S , I_D would probably increase at a decreasing rate with increases in the scale of the trial. Unlike the process of trialing for skill development, trialing for reduced uncertainty can lead to a *lower* estimate of the profitability of the crop. In such cases it does not mean that the information has a negative value, since the reduction in planted area which results is a better decision.

4.3. Risk attitudes affect adoption

If the subjective expected utility model of risk attitudes is accepted, the inclusion of risk attitudes in the dynamic adoption model is, in principle, straight forward. Instead of assuming that the farmer maximizes $E(NPV)$, we now have a model where the farmer maximizes $E(U[NPV])$. Within this modified adoption model I_S and I_D affect the distribution of NPV which is used to calculate $E(U[NPV])$. From here, assuming a well-behaved utility function, we can proceed to solve this adoption model to find the optimal investment in the innovation if the farmer chose to trial in the first year, evaluated using $E(U[NPV])$ as the objective, rather than $E(NPV)$.

The consequences of risk aversion for adoption are not unambiguous, since they depend on farmers' perceptions of the relative riskiness of old and new technologies and the levels of uncertainty faced. Adoption of new technology might either reduce or increase risk in the long run. Perhaps the cases of reduced risk explain the occasional empirical evidence for a positive relationship between risk aversion and adoption (e.g. Shapiro et al., 1992). On the other hand, uncertainty would tend to always be greater for the new technology than for the old, and so would discourage adoption by risk-averse farmers. Similarly, if a lack of experience with use of a new technology increases the risk of implementation failure, risk aversion would again discourage adoption. Overall, a negative influence of risk aversion on adoption has been the more common empirical finding (e.g. Brink and McCarl, 1978; Marra and Carlson, 1990; Abadi Ghadim 2000).

5. Research needs and emerging issues in risk and technology adoption

5.1. Crop biotechnology

Crop biotechnology has only been commercially available for, at most, five years. The adoption curves for some of these technologies, especially the herbicide resistant crop types, have been steep, even though these crops have not yet been approved for animal feeding or human consumption in many parts of the world (Table 2).

For transgenic crops with herbicide tolerance, such as Roundup Ready[®] (RR) soybeans, there seemed to be initial uncertainties about relative profitability compared to conventional weed control systems. Initially, the transgenic crops had lower yields in some varieties and locations than their conventional counterparts. Nevertheless, RR

soybeans are the most widely adopted of the transgenic crops to date. For insect resistant varieties, the uncertainty comes primarily from variable pest infestations. (For example Bt corn faces variable European corn borer densities that are linked to location, insect egg laying weather, and time of planting.) Bt cotton uncertainty depends upon insect infestation levels, the mixture of pest species, and whether the pests in a given location are resistant to conventional insecticides. Marketing costs

Table 2

Transgenic crops by state and US total, percent of planted crop acres, 2000

Crop/state	Transgene type			
	Herbicide resistant	Insect resistant	Stacked gene	All transgenic varieties
<i>Soybeans</i>				
AR	43			43
IL	44			44
IN	63			63
IA	59			59
KS	66			66
MI	50			50
MN	46			46
MS	48			48
MO	62			62
NE	72			72
ND	22			22
OH	48			48
SD	68			68
WI	51			51
Other soybean states				
USA	54			54
	54			54
<i>Corn</i>				
IL IN3		13	1	17
IA	4	7		11
KS	5	23	2	30
MI	7	25	1	33
MN	4	8		12
MO	7	28		37
NE	6	20	2	28
OH	8	24	2	34
SD	3	6	2	9
WI	11	35	2	48
Other corn states	4	13	1	18
USA	6	10	1	17
	6	18	1	25
<i>Cotton</i>				
AR	23	33	14	70
CA	17	3	4	24
GA	32	18	32	82
LA	13	37	30	80
MI	13	29	36	78
NC	29	11	36	76
TX	33	7	6	46
Other cotton states	21	17	36	74
USA	26	15	20	61

Source: Agricultural Statistics Board, NASS, USDA.

and prices of transgenic crops are also random variables, especially with identity preservation requirements and uncertain premiums for products not containing transgenic crop ingredients.

In addition to the Marra et al. (in press) study of Bt cotton adoption discussed previously, there are only a few adoption studies to date of this important new technology. The simulation study of Bt corn adoption by Hyde et al. (1999) is based on data from expert opinions concerning cornborer infestations in Indiana. They study found that mean profitability estimates varied systematically with European Corn Borer (ECB) infestation levels. They found that high levels of absolute risk aversion could make Bt corn attractive under certain circumstances.

Darr and Chern (2000) surveyed Ohio growers about their adoption of Bt corn and RR soybeans in 1998–2000. There are consistent findings of higher adoption rates in 2000 for farmers with more acres and those in higher ECB infestation areas. They found higher probabilities and intensities of adoption with farmers who routinely “sell grain forward”, a proxy for risk aversion. Farmers who said they were “concerned about price premiums” for conventional corn had lower adoption intensities. None of these “risk aversion measures” was found to be associated with changes in the adoption or adoption intensity of RR soybeans. These indirect measures of risk aversion may change over time, or be correlated with unobservable factors. This can result in unstable and biased parameter estimates in these adoption models. Elicitation of subjective risk preferences and price and yield distributions, while still an art, could alleviate some of the statistical problems.

5.2. *Precision agriculture*

The decision problem on whether to adopt precision agriculture technologies involves the usual risk-related considerations reviewed earlier, but additionally highlights the issue of spatial variability of productivity within a farm. Bennett and Pannell (1998) developed a framework for evaluating a precision agriculture technology intended to control the spatial distribution of a crop input (a herbicide) in response to a spatially variable biophysical parameter (the density of weeds). They noted that the structure of the decision problem for precision agriculture is well represented in a Bayesian decision theory framework. The precision technology provides information that reduces uncertainty about the net benefits of applying an input at a particular location. The information allows an improved, but still imperfect, “decision” to be made by the input applying technology.

In an application of the framework to a precision weed management technology, Pannell and Bennett (1999) found that their comprehensive representation of the problem made a substantial difference to results of the evaluation. A simple analysis based only on input savings produced results very favorable to the technology, but a more complete representation of the problem¹ resulted in a distinctly

¹ The additional complexities represented were: increased weed competition in those patches which are left unsprayed; failure to spray high weed density patches in some cases; falsely spraying low weed density patches in some cases; and allowance for the fact that the best alternative spraying strategy (other than the precision sprayer) will vary in different circumstances.

adverse conclusion. This suggests that an accurate representation of the risk-related aspects of precision agriculture technologies may be critical in accurately evaluating their potential. One might hypothesize that, given the difficulty and complexity of correctly evaluating precision technologies, and their apparent attractiveness when considered from a simple perspective, farmers might tend to over-adopt some precision technologies relative to their true value. On the other hand, the fixed costs of some precision farming technology components may result in investment delays relative to predictions of the neo-classical model.

5.3. *Environmental management technologies*

The few studies which have examined the importance to the adoption process of on-farm trialing of new technologies have concluded that it does play a key role (e.g. Abadi Ghadim, 2000; Marra et al., in press). If it is true that small-scale on-farm trialing is a necessary step in the adoption process; there are important and concerning implications for the adoption of technologies for some environmental problems. For example, Pannell (in press) examined the case of establishing perennial plants in place of traditional annual crops in Australia, in order to control groundwater levels in regions at risk of dryland salinization. He identified a wide range of difficulties in the process of trialing to evaluate adoption:

- observability of treatment impacts on groundwater levels is low or observations are costly;
- there can be long time lags between treatment and effect;
- in a common property groundwater problem, the effectiveness of a local trial by an individual farmer may be compromised by non-trialing neighbors;
- the necessary scale of implementation for effectiveness against salinity is high, so that a small scale trial gives poor information about the effectiveness of full adoption;
- the quality of implementation is likely to be lower for technologies (like perennials) which are outside the normal range of experience for most farmers;
- resources required for trialing pose an impediment, especially where large-scale “trials” are needed; and
- risks of trial failure are likely to be higher for very unfamiliar technologies, reducing the incentive to conduct a trial.

Similarly, Pannell and Zilberman (in press) identified trialability and observability as likely impediments to the adoption of “integrated weed management” practices to delay the onset of herbicide resistance. They concluded, “rapid adoption of integrated weed management systems, involving combinations of unfamiliar, complex, and expensive treatments that are difficult to trial, is unlikely to occur until it is essential.” These observations provide a potentially valuable insight into the adoption problem for many environmental management technologies. It may be productive to re-examine apparently unexplained cases of non-adoption of environmental management technologies through the lens of their trialability, as well as their “irreversibility” characteristics.

Another environmental management issue currently of great concern in parts of the USA is the management of livestock waste from confined animal feeding operations. The open-pit lagoon technologies currently used in southeastern US swine production, for example, have been blamed, at least in part, for reduced water quality and fish kills in nearby rivers and estuaries. Several major lawsuits have been filed by environmental groups and other interested parties against the North Carolina swine industry. New, more environmentally friendly, technologies are in the early stages of development and are likely to experience adoption lags from the aspects of uncertainty discussed throughout this section. There is a real opportunity for economic analysis to shed some light on the potential adoption process, as well as predicting the relative efficiency of the performance of various regulations and incentives proposed by government agencies.

6. Concluding comments

Relatively recent research about the roles of risk, uncertainty and learning in the adoption of agricultural technologies has finally provided compelling support for the long-held and often-stated view that adoption processes are strongly affected by risk-related issues. We have noted the importance of distinguishing between different aspects of risk, uncertainty and learning in correctly understanding the adoption process. These aspects include: farmers' perceptions about the riskiness of a technology, farmers' attitudes to risk, the role of trialing and learning in adoption decisions, and the option value of delaying adoption. There is considerable scope for wider application of the more comprehensive and rigorous research methods used in some of the studies reviewed here. We have also highlighted a number of emerging issues for which further high quality research, which properly recognizes the roles of risk, uncertainty and learning, appears likely to result in better understanding and explanation of agricultural technology adoption.

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